

Clinical Risk Assessment Report: Patient 51

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: Borderline / Low Confidence
- Absolute Predicted Risk: 24.3% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.10x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which elevated the patient's absolute risk by +4.2%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Clinical Sensitivity Analysis (Borderline Patient):

PREDICTION INSTABILITY WARNING: This patient's risk profile sits precisely on the borderline. Consequently, the model's prediction is highly sensitive. Minor biological variations or measurement errors in the following features would flip the final clinical prediction:

- Large Zone Emphasis (CT): An increase would alter the prediction outcome.
- Patient Age: An increase would alter the prediction outcome.
- Small Zone Emphasis (PET): A decrease would alter the prediction outcome.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.32 [Bottom 20%] (+4.2%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Patient Age (Age) **Value: 52.0 [Bottom 15%] (+4.2%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 1.29 [Top 9%] (+2.1%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT) **Value: -2.03 [Bottom 4%] (+0.9%)**
Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

Tumor Sphericity (PET) (SHAPE_Sphericity(onlyFor3DROI)).PET) **Value: 0.96 [Top 13%] (+0.9%)**
Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

3. Protective / Mitigating Factors (Reducing Risk)

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 1.53 [Top 4%] (-2.8%)**
Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

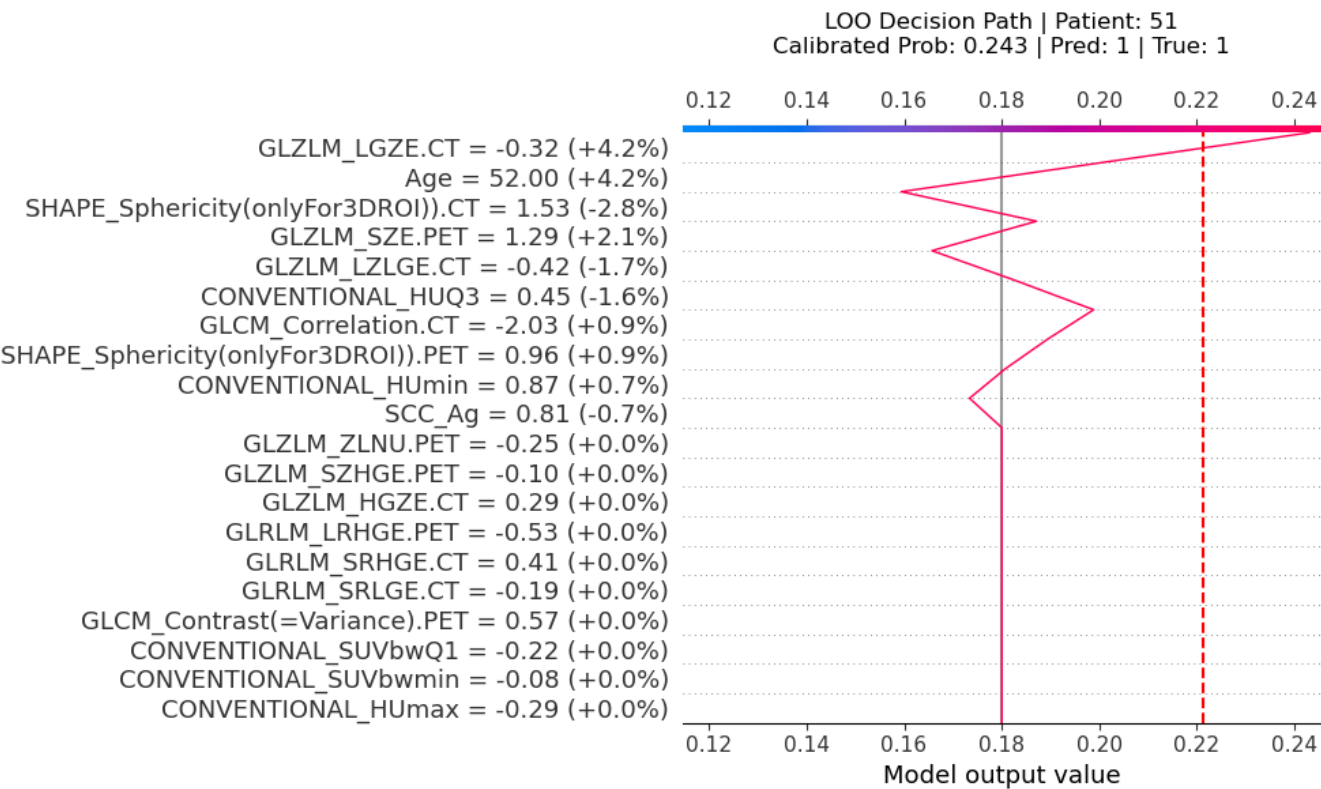
Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.42 [Bottom <1%] (-1.7%)**

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.45 [Avg (57th %ile)] (-1.6%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.

